F FWallSim · Firewall Course

CLASS 20

The branch can reach the tunnel, but not HQ

Troubleshooting: return route, reverse-path check

Advanced | ~15 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

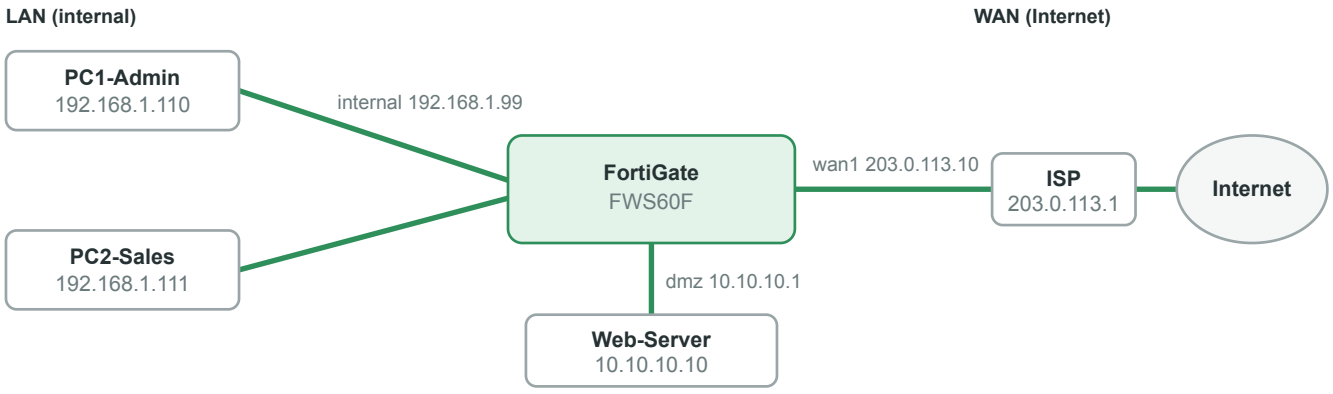
YOU WILL LEARN

- Troubleshooting: return route
- reverse-path check

BEFORE YOU START

Finish Class 19 first.

The lab network



The goal

THE IDEA

Branch-PC cannot RDP to PC1. The tunnel is up and the Branch-to-HQ policy is correct, yet every packet is dropped. Someone "fixed" the branch route last week.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **The branch can reach the tunnel, but not HQ** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **Branch-PC can RDP to PC1**

   In the Traffic Generator send Branch-PC → 192.168.1.110, RDP (TCP/3389) and read where it is dropped. After the fix, send it again.

2. **PC1 can ping Branch-PC**

   Test the reverse direction: PC1-Admin → 10.20.0.10, Ping (quick test "PC1 pings Branch-PC").

Hints

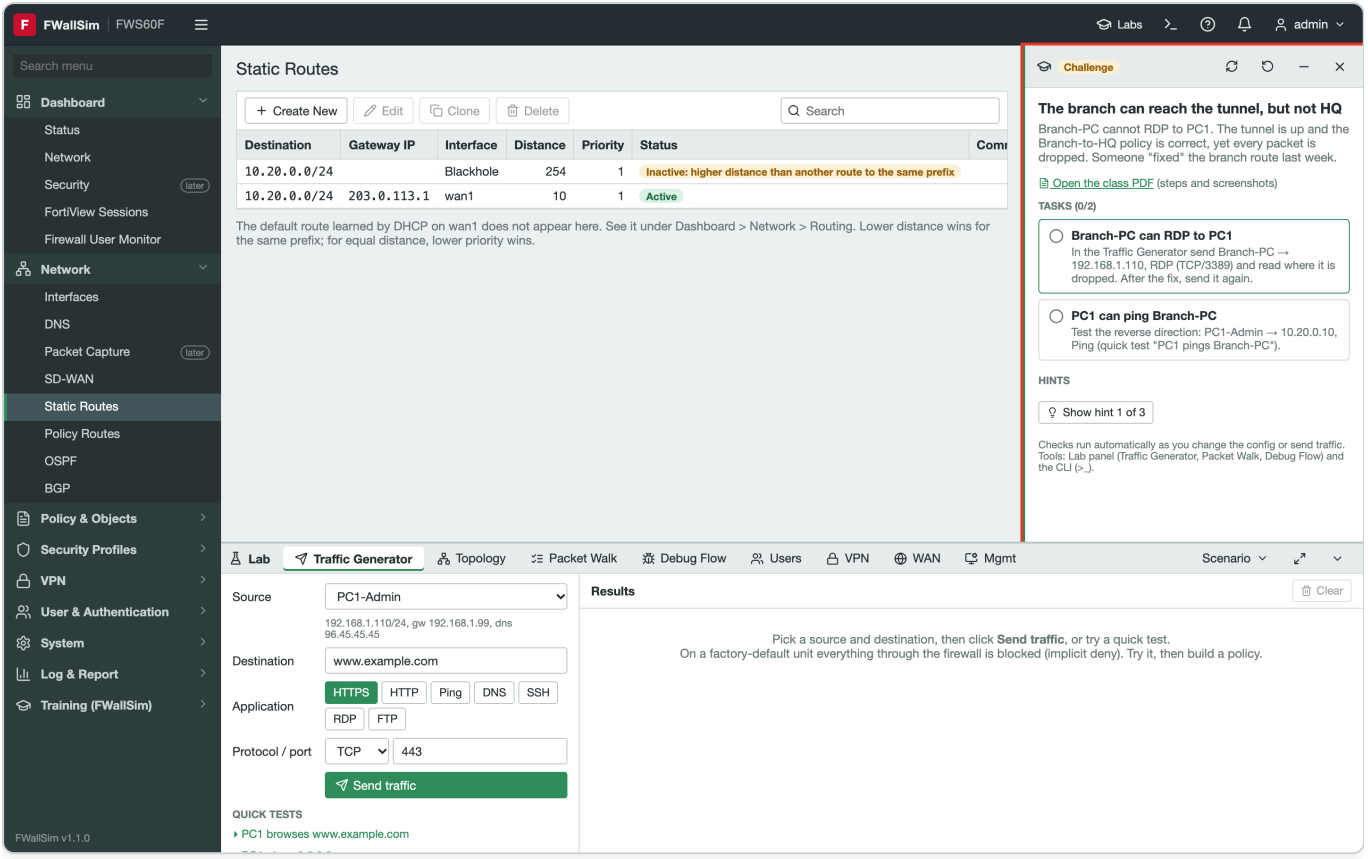
1. Look at the Packet Walk (or "diagnose debug flow" in the CLI): it says "reverse path check fail, drop".
2. Which route does the FortiGate use to reach 10.20.0.10? Check Dashboard > Network > Routing or "get router info routing-table all".
3. The route back to the branch must point into the tunnel, not out to the internet.

Solution

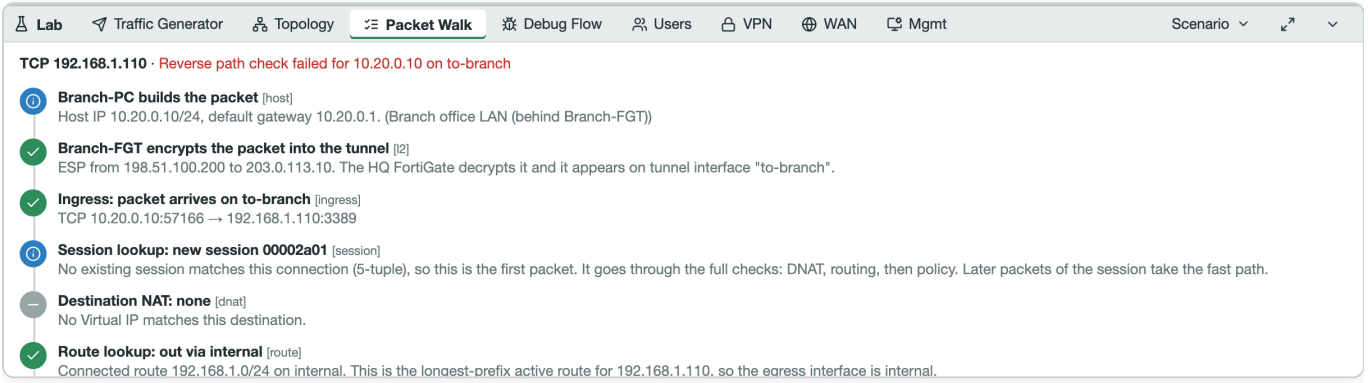
1. Network > Static Routes > edit the 10.20.0.0/24 route that uses wan1.
2. Interface: to-branch (no gateway).
3. OK, then test both directions.

Screenshots

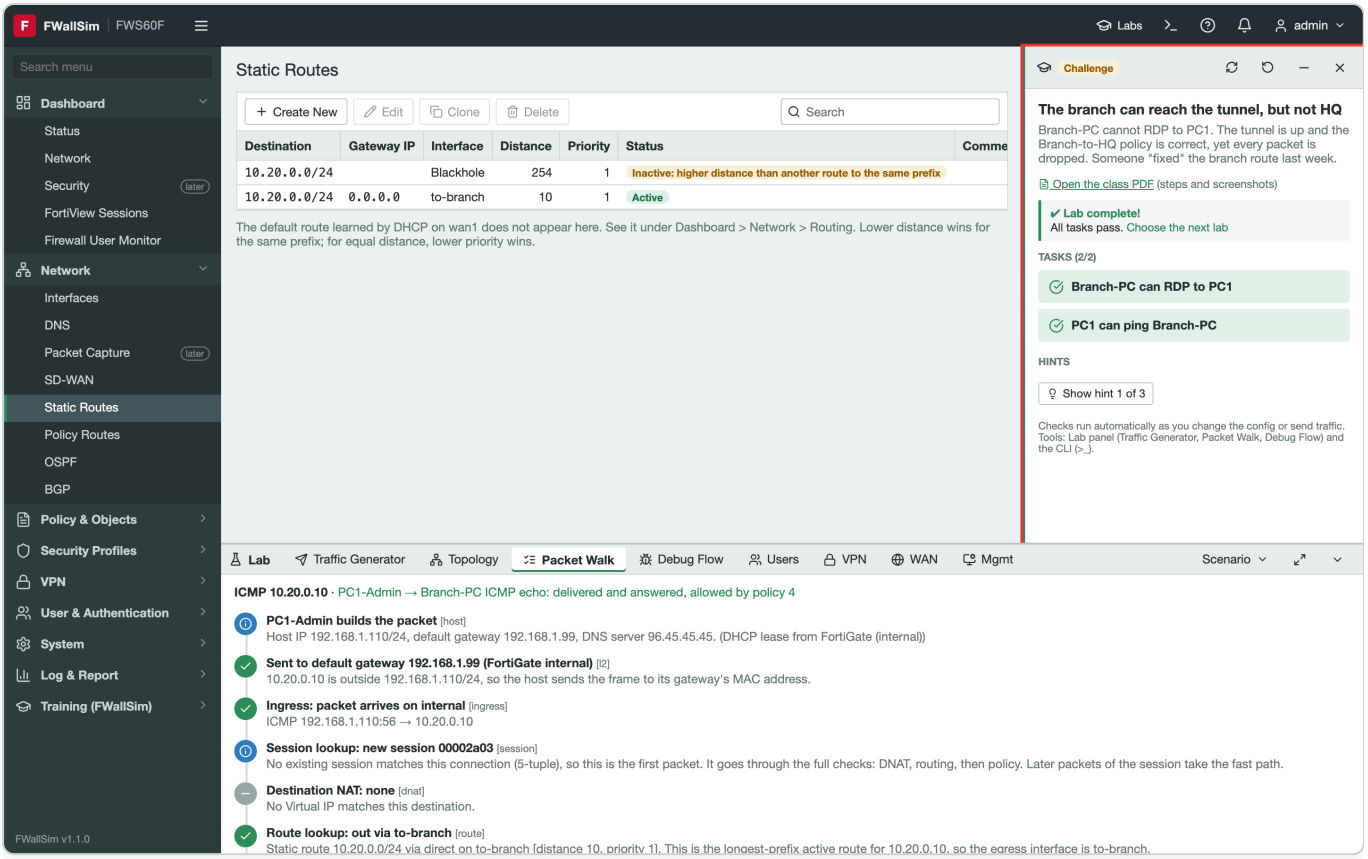
Numbered red markers show where to click or what to look at.



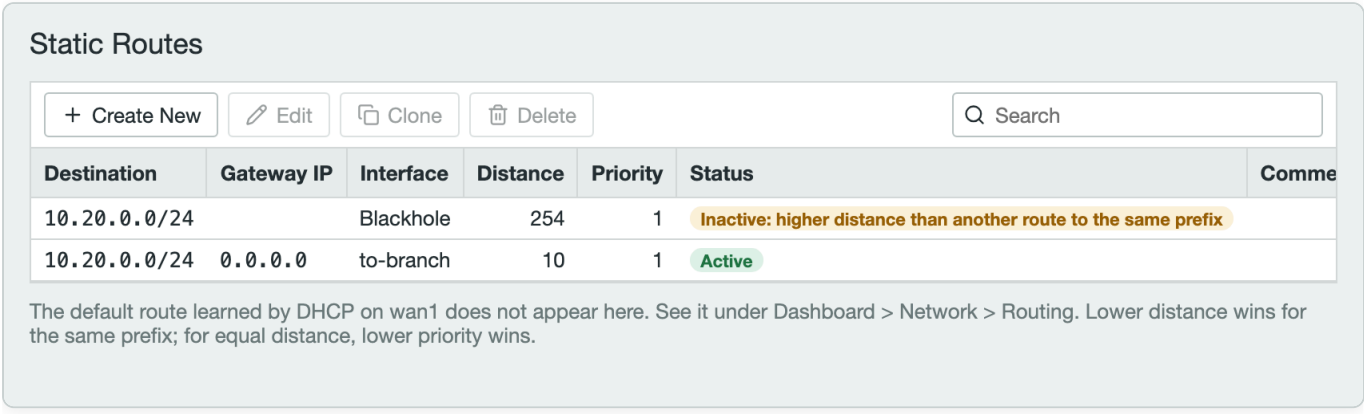
Lab Started



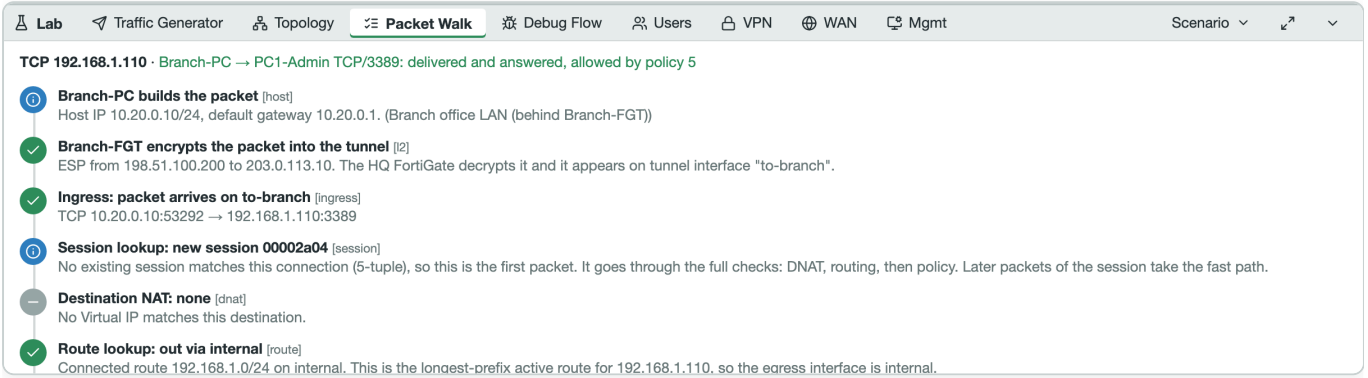
Packet Walk Before



Lab Solved



Static Routes



Packet Walk After